

AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MS102

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 14/Apr. /2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 1.5hrs

Instructions:

- Total Marks: **30**
- This exam consists of 2 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART I: Answer the entire Multiple Choice Questions (MCQ). [15 Marks]

1. A particle moves along y-axis with position y= 5t^3 - 2t^2 + t. Its velocity at t = 2 s is;
A. 36 m/s B. 53 m/s C. 28 m/s D. 30 m/s

2. Two objects of mass 3 kg and 6 kg moving in opposite directions with speeds 4 m/s and 2 m/s collide in-elastically. Velocity of combined mass after collision is
A. 0 m/s B. 1 m/s C. 2 m/s D. 3 m/s

3. The ratio of kinetic energy to potential energy of a freely falling body after falling half the height h is
A. 1:1 B. 1:2 C. 2:1 D. 3:1

4. A particle has velocity components Vx = 3 m/s, vy = 4 m/s. Magnitude of velocity is
A. 5 m/s B. 7 m/s C. 6 m/s D. 25 m/s

5. The slope of a displacement-time graph for uniform acceleration is
A. Linear B. Quadratic C. Constant D. Zero

6. A car started from rest to 20 m/s in 2 second. Acceleration is
A. 8 m/s^2 B. 6 m/s^2 C. 10 m/s^2 D. 12 m/s^2

7. A particle is moving in straight line with velocity $v = 4t^2$. Acceleration after 3 s is
A. 12 m/s^2 B. 24 m/s^2 C. 8 m/s^2 D. 16 m/s^2

8. Newton's second law can be written as
A. F = ma B. F = mv C. F = m/v D. F = m + a

9. A body of mass 0.5 kg moving at 10 m/s collides with a wall and rebounds with 8 m/s. Impulse magnitude experienced is
A. 9 Ns B. 10 Ns C. 8 Ns D. 18 Ns

10. A stone is thrown vertically upward. At the highest point:
A. Velocity = g, acceleration = 0 B. Velocity = 0, acceleration = g
C. Velocity = 0, acceleration = 0 D. Velocity = g, acceleration = g

11. If the magnitude of two perpendicular vectors are 6 and 8, the magnitude of resultant=
A. 10 B. 14 C. $\pm\sqrt{100}$ D. 12

12. A particle moves under uniform acceleration covering distances of 5 m and 9 m in 1st and 2nd seconds. Acceleration is
A. 4 m/s^2 B. 2 m/s^2 C. 3 m/s^2 D. 5 m/s^2

13. The base quantities among the following are
A. Mass, Length, Force B. Mass, Time, Velocity
C. Mass, Length, Time D. Force, Length, Time

14. Work done by a force perpendicular to displacement is
A. Maximum B. Zero C. Positive D. Negative

15. A body of mass 4 kg moving at 5 m/s collides elastically with a stationary 6 kg body. After collision, velocity of 4 kg body return is

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A. 1 m/s B. 2 m/s C. 3 m/s D. 4 m/s

16. Distance covered in n^{th} second under constant acceleration a from rest is given by
A. a(2n-1)/2 B. a n C. a n^2 D. 1/2 a n^2

17. If F = 10 N acts at 60° to displacement 5 m, work done is
A. 25 J B. 50 J C. 10 J D. 5 J

18. A block on an inclined plane accelerates at 3 m/s^2. Gravity component along the incline is 5 m/s^2. Friction acceleration is
A. 2 m/s^2 B. 8 m/s^2 C. 1 m/s^2 D. 0 m/s^2

19. Momentum of a body is doubled while mass remains constant. Kinetic energy is
A. Doubled B. Quadrupled C. Halved D. Same

20. The force required to keep a 2 kg mass moving at acceleration 4 m/s^2 is
A. 8 N B. 4 N C. 2 N D. 16 N

21. A particle moves along x-axis with position x = 3t^2 + 2t + 1 (SI units). Its acceleration at t = 2 s is
A. 3 m/s^2 B. 6 m/s^2 C. 12 m/s^2 D. 8 m/s^2

22. If the mass of a body doubles and its velocity halves, its momentum changes by
A. Remains the same B. Halves C. Doubles D. Quadruples

23. The dimensional formula of gravitational constant G is
A. [M^-1 L^3 T^-2] B. [M L^-3 T^2] C. [M L^2 T^-2] D. [M^-1 L^2 T^-2]

24. Two vectors of magnitude 5 and 12 are opposite. The magnitude of their resultant is
A. 13 B. 17 C. 7 D. 10

25. A car accelerates uniformly from 10 m/s to 30 m/s in 20 s. Distance covered is
A. 400 m B. 500 m C. 600 m D. 700 m

26. A force F = 10 N acts on a mass of 2 kg for 3 s. The impulse imparted is
A. 10 Ns B. 20 Ns C. 30 Ns D. 60 Ns

27. A block slides down a frictionless incline of height h. Its speed at the bottom is
A. $\sqrt{(2gh)}$ B. $\sqrt{(gh/2)}$ C. gh D. 2gh

28. Two forces act on a particle: F1 = 3i + 4j N, F2 = -i + 2j N. Net force magnitude is
A. 5.25 N B. 6.32 N C. 4 N D. 7 N

29. A particle is dropped from rest. The distance covered in the nth second is given by
A. u + 1/2 g(2n-1) B. 1/2 g(n^2) C. g(n^2) D. gn

30. A car moves with uniform acceleration. Its velocity-time graph is
A. Horizontal line
B. Straight line with positive slope
C. Parabola
D. Exponential curve

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Part II: Answer the following questions. [15 Marks]

Question 1: (8 Marks)
A car of mass 1200 kg accelerates from 0 to 30 m/s in 10 s. Friction and air resistance produce 800 N. Find:
(a) Force applied by engine
(b) Work done by engine

Question2 (7 marks)
A 0.2 kg bullet moving at 500 m/s embeds into a 1.8 kg wooden block at rest. Find:
(a) Velocity of block after collision
(b) Impulse on the block

End of questions

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